

Laxmi Charitable Trust's
 Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical :-IX

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Class :- TYIT	Batch :- 2
Date of Assignment :- 18/08/2026	Date/Time of Submission :- 18/08/2026

A) Develop CRUD operations using [ASP.NET](#) Core Web API and Entity Framework Core.

Code:-

Student.cs:-

```
using Microsoft.EntityFrameworkCore;
using StudentAPI.Models;
namespace StudentAPI.Data
public class Student : DbContext
{
    public Student(DbContextOptions<Student> options: base(options)
    {
    }
    public DbSet<Student> Students { get; set; }
}
```

App.DbContext:-

```
using Microsoft.EntityFrameworkCore;
using StudentAPI.Models;
namespace StudentAPI.Data
public class Student : DbContext
{
    public Student(DbContextOptions<Student> options: base(options)
    {
    }
    } public DbSet<Student> Students { get; set; }
}
```

AppSetting.json:-

```
using Microsoft.EntityFrameworkCore;
using StudentAPI.Models;
namespace StudentAPI.Data
public class Student : DbContext
{
    public Student(DbContextOptions<Student> options: base(options)
    {
    }
    } public DbSet<Student> Students { get; set; }
}
```

Program.cs:-

```

using Microsoft.EntityFrameworkCore;
using StudentAPI.Data;
var builder = WebApplication.CreateBuilder(args);
builder.Services.AddControllers();
builder.Services.AddDbContext<Student>(options =>
    options.UseSqlServer(
        builder.Configuration.GetConnectionString("DefaultConnection")));
builder.Services.AddEndpointsApiExplorer();
builder.Services.AddSwaggerGen();
var app = builder.Build();
app.UseSwagger();
app.UseSwaggerUI();
app.UseHttpsRedirection();
app.UseAuthorization();
app.MapControllers();
app.Run();

```

StudentController.cs:-

```

using Microsoft.AspNetCore.Mvc;
using StudentAPI.Data;
using StudentAPI.Models;
namespace StudentAPI.Controllers
{
    [Route("api/[controller]")]
    [ApiController]
    public class StudentsController : ControllerBase
    {
        private readonly Student _context;

        public StudentsController(Student context)
        {
            _context = context;
        }
        // GET ALL
        [HttpGet]
        public IActionResult GetStudents()
        {
            return Ok(_context.Students.ToList());
        }
        // GET BY ID
        [HttpGet("{id}")]
        public IActionResult GetStudent(int id)

```

```

{
    var student = _context.Students.Find(id);
    if (student == null)
        return NotFound();
    return Ok(student);
}
// INSERT
[HttpPost]
public IActionResult AddStudent(Student student)
{
    _context.Students.Add(student);
    _context.SaveChanges();
    return Ok(student);
}
// UPDATE
[HttpPut("{id}")]
public IActionResult UpdateStudent(int id, Student student)
{
    var data = _context.Students.Find(id);
    if (data == null)
        return NotFound();
    data.Name = student.Name;
    data.Age = student.Age;
    data.Course = student.Course;
    _context.SaveChanges();
    return Ok(data);
}
// DELETE
[HttpDelete("{id}")]
public IActionResult DeleteStudent(int id)
{
    var student = _context.Students.Find(id);
    if (student == null)
        return NotFound();
    _context.Students.Remove(student);
    _context.SaveChanges();
    return Ok();
}
}

```

Output:-

POST/api/Students

Parameters

No parameters

Request body

Edit Value | Schema

```
{
  "name": "Nidhi",
  "age": 19,
  "course": "BSCIT"
}
```

Code

Details

200

Response body

```
{
  "id": 1,
  "name": "Nidhi",
  "age": 19,
  "course": "BSCIT"
}
```

Response headers

```
content-type: application/json; charset=utf-8
date: Tue, 18 Aug 2026 06:47:44 GMT
server: Microsoft-IIS/10.0
x-powered-by: ASP.NET
```

PUT/api/Students/{id}

Parameters

Name	Description
id * required	

integer(\$int32)1(path)

Request body

Edit Value | Schema

```
{
  "id": 0,
  "name": "Nidhi Yadav",
  "age": 19,
  "course": "IT"
}
```

Code

Details

200

Response body

```
{
  "id": 1,
  "name": "Nidhi Yadav",
  "age": 19,
  "course": "IT"
}
```

Response headers

```
content-type: application/json; charset=utf-8
date: Tue, 18 Aug 2026 06:56:59 GMT
server: Microsoft-IIS/10.0
x-powered-by: ASP.NET
```

DELETE/api/Students/{id}

Parameters

Name	Description
id * required	

integer(\$int32)1(path)

Curl

```
curl -X 'DELETE' \
'https://localhost:44386/api/Students/1' \
-H 'accept: */*'
```

Request URL

https://localhost:44386/api/Students/1

Server response

Code

Details

200

Response headers

```
date: Tue, 18 Aug 2026 07:00:34 GMT
server: Microsoft-IIS/10.0
x-powered-by: ASP.NET
```

	Id	Name	Age	Course
▶	2	Nidhi	20	BSCIT
⚙	NULL	NULL	NULL	NULL

StudentAPIWithEF1.0OAS 3.0

https://localhost:44386/swagger/v1/swagger.json

Students

GET/api/Students

POST/api/Students

GET/api/Students/{id}

PUT/api/Students/{id}

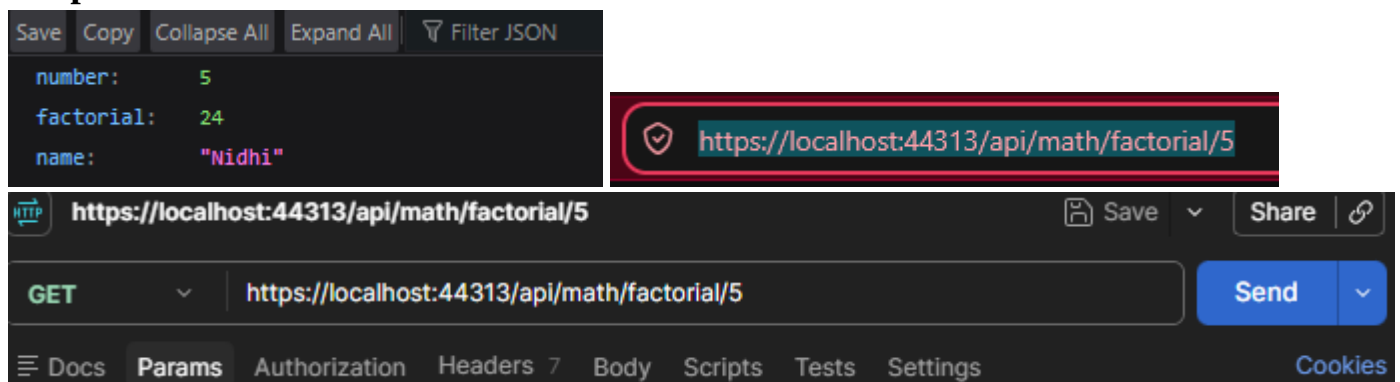
DELETE/api/Students/{id}

B) Create an API endpoint to calculate Factorial of a Number and test it using Postman.

Code:-

```
using Microsoft.AspNetCore.Mvc;
namespace FactorialAPI.Controllers
{
    [Route("api/[controller]")]
    [ApiController]
    public class MathController : ControllerBase
    {
        [HttpGet("factorial/{number}")]
        public IActionResult GetFactorial(int number)
        {
            if (number < 0)
            {
                return BadRequest("Factorial not possible for negative numbers.");
            }
            long factorial = 1;
            for (int i = 1; i < number; i++)
            {
                factorial *= i;
            }
            return Ok(new
            {
                Number = number,
                Factorial = factorial,
                Name = "Nidhi"
            });
        }
    }
}
```

Output:-



The screenshot displays the Postman interface. On the left, a JSON response is shown with the following data:

```
{
  "number": 5,
  "factorial": 24,
  "name": "Nidhi"
}
```

On the right, the URL bar shows the request: `https://localhost:44313/api/math/factorial/5`. Below the URL bar, the HTTP method is set to `GET`, and the same URL is entered in the request field. The `Send` button is visible. At the bottom, the `Params` tab is selected, showing no parameters.

Body ↕

{ } JSON ↕

▶ Preview

🖼️ Visualize

1 {

2 "number": 5,

3 "factorial": 24,

4 "name": "Nidhi"

5 }